AXIOM: Correctness-Aware Multi-Agent Slice Admission Control with Federated Training

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Abstract

Fifth generation (5G) network slicing must admit heterogeneous service requests across many cooperating base stations under strict per-slice guarantees. This paper presents AXIOM, a multi-agent slice admission controller built on one shared cluster encoder that feeds a small decision head per slice. We evaluate it under a pre-registered protocol in two simulators, one with static load and a harder one with time-varying load and inter-cell coupling, against single-agent and independent deep Q-networks (DQNs), multi-agent proximal policy optimisation (MAPPO), and tuned threshold rules. AXIOM's advantage over the baselines grows with the size of the cluster. At 7 and 19 gNodeBs (gNBs) it keeps Ultra-Reliable Low-Latency Communication (URLLC) and enhanced Mobile Broadband (eMBB) Service Level Agreement (SLA) compliance near 97%, while every baseline stays below 84%, with 7,782 parameters at any cluster size. At three gNBs it reaches 98.9% of an oracle rule's reward under static load, and under coupling it beats the static rule by 39% and leads every learned baseline. Training AXIOM in federated form adds no significant cost in reward. The gains have clear bounds, since a three-parameter rule tuned with domain knowledge outperforms every learned policy under coupling, a single DQN over the joint state can match AXIOM in some settings, and graph attention adds nothing over a per-node encoder. Independent recomputation and pre-stated checks on fresh seeds confirm the main findings. A live OpenAirInterface (OAI) testbed validates the control path on an Open Radio Access Network (O-RAN) stack, while AXIOM itself is evaluated offline.

Keywords: Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

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1 Introduction

Network slicing virtualises the physical radio access network (RAN) to serve heterogeneous services under Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) categories. Each slice carries distinct Service Level Agreement (SLA) targets that are in turn based on Quality of Service (QoS) thresholds. Our earlier work [16, 17] validated that a Deep Q-Network (DQN) admission controller can enforce SLAs for a single gNodeB (gNB) in a live OpenAirInterface (OAI) testbed. However, a real deployment spans multiple gNBs that share backhaul, spectrum planning, and a cluster-wide load-balance objective. Thus, slice admission at one gNB is rarely isolated, and coordinated admission is in fact a multi-gNB, multi-agent challenge. This research aims to address whether a shared, graph-structured view of the cluster could enable per-gNB agents to decide better than a single agent on a flattened state, as opposed to independent agents. Fig. 1 places the research premise in context.

This paper introduces AXIOM, an Adaptive Cross-cell Integrity-aware O-RAN Multi-agent slice-admission controller, where O-RAN denotes the Open Radio Access Network. AXIOM is a centralised-training, decentralised-execution multi-agent architecture that is built on a shared Graph Attention Network (GAT) [38] encoder over a multi-gNB topology, which feeds decentralised per-slice Q-heads. We deliver it together with the evaluation discipline needed to trust what it reports. Three gaps from our systematic review of 67 core studies motivate this pairing, as Section 2 details. First, no reviewed study checks whether its own outcome metric can score a degenerate policy as well as a correct one. Second, none replicates a headline architectural finding on an independent seed sample. Third, none tests whether a live testbed can reach

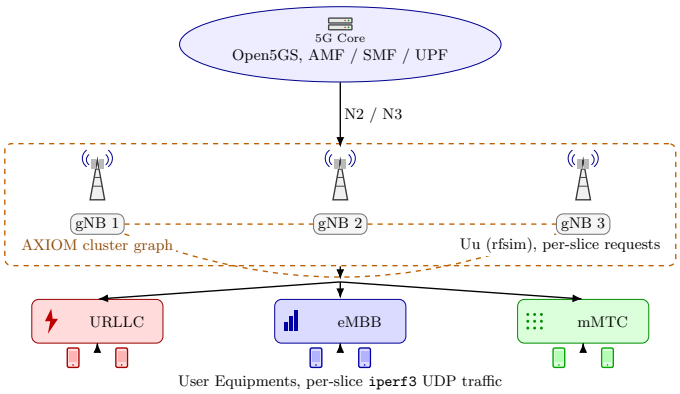


Fig. 1 Conceptual system view. A cluster of gNBs, coordinated by the shared AXIOM encoder and backhauled to a 5G core, serves UEs across three slices. AXIOM is evaluated offline; the live rig runs one gNB.

the contention regime its offline claims assume. We address these gaps through a per-component realisation of the architecture, namely the GAT encoder, the Multi-Agent Reinforcement Learning (MARL) controller, the Federated Learning (FL) coordinator, and the Differential Privacy (DP) layer, each measured and isolated rather than folded into one end-to-end number. We apply a correctness-aware metric pair and a reproduction-versus-replication discipline throughout, and we add a live single-gNB deployment that tests, rather than assumes, whether the rig reaches genuine per-slice scarcity.

We evaluate AXIOM in two offline simulators of a gNB cluster. The first, Env A, has a static load in which shedding every mMTC request is optimal. The second, Env B, adds time-varying load and inter-cell coupling, so that no fixed rule is optimal and neighbouring gNBs affect each other. We compare AXIOM with single-agent and independent deep Q-networks, with multi-agent proximal policy optimisation (MAPPO), and with tuned threshold rules. We also extend it to federated training with differential privacy, stress it under gNB dropout, demand spikes, and agent churn, and grow the cluster to 7 and 19 gNBs under fully-connected, ring, and hex-grid adjacency. The contributions of this paper are as follows.

- AXIOM, a multi-agent controller with a shared cluster encoder and decentralised per-slice Q-heads, whose parameter count does not grow with the cluster. In Env A it keeps priority-slice compliance near 97% at 7 and 19 gNBs, where both DQN baselines and MAPPO stay below 84%, and in Env B it leads every learned baseline at three gNBs.
- A harder environment with time-varying load and inter-cell coupling, in which learned policies beat static rules and a tuned threshold rule sets the bar they have yet to reach.
- An evaluation protocol for learned slice admission control that combines trivial-policy checks, per-slice correctness metrics, independent-seed replication, tuned rule baselines, and a pre-registered analysis. It caught seven evaluation failures in this work, and every main finding was then confirmed on fresh seeds.
- Clear bounds on the architecture's claims, namely that graph attention adds nothing over a per-node encoder and that per-step differential privacy gives only a weak guarantee.

2 Related Work

Our prior systematic review [18] screened 6,286 records down to 67 core studies from 2024 to 2026 on machine-learning-driven, user-centric distributed network slicing. The review organised them around three enabling pillars, namely topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation. Each pillar is well developed on its own, but the three are rarely combined under realistic constraints, and never with a validated live deployment. No reviewed study checks whether its own evaluation metric can mask a degenerate policy, and none replicates a headline finding on an independent seed sample, which is the gap this paper fills.

Studies from the review's core sample show the state of each pillar. For admission control, Ojijo et al. [24] admit slice requests with a sequential twin actor-critic over a multi-dimensional state space and a distributed action space, and Tao et al. [34] use a digital twin to accelerate online deep reinforcement learning for admission. For multi-agent control, Doanis and Spyropoulos [8] make multi-agent DQNs sample-efficient for multi-domain inter-slice orchestration, Li et al. [20] propose a scalable multi-agent framework for dynamic RAN slicing, and ICE-CREAM [25] coordinates fully cooperative decentralised agents for energy-efficient RAN slicing. For graph representation, Hao et al. [11] combine a graph neural network with meta-learning to make bandwidth allocation scalable and transferable. For federation, Amiri et al. [4] split virtual network functions in O-RAN slicing with federated deep reinforcement learning, and Ming et al. [22] apply federated deep reinforcement learning to prediction-based slice mobility. Closest to our evaluation concern, Tsampazi et al. [37] design and evaluate many deep reinforcement learning agents for O-RAN in an automated pipeline, a systematic comparison that our trivial-policy and independent-seed checks extend.

AXIOM also sits inside the broader O-RAN effort to run learned control loops as applications over standard interfaces. Polese et al. [27] survey that architecture and its open problems, while ColO-RAN [26] and Tsampazi et al. [36] train and compare deep reinforcement learning controllers for slicing on large experimental platforms. Learned slice admission predates O-RAN, from market-driven admission of slice requests [5] to reinforcement learning for RAN slicing [30]. For cooperating agents, value factorisation such as QMIX [29] and centralised-critic policy gradients such as MAPPO [40] are the standard training schemes, and we compare against MAPPO directly. Constrained multi-agent learning has also been applied to Open RAN slicing at scale [41], and Khani et al. [13] survey deep reinforcement learning for slice admission in cloud RAN. Within this journal, recent work applies actor-critic learning to joint resource allocation in RAN slicing [15], double deep Q-networks to next-generation networks [12], and delay-aware slice allocation to cloud RAN [3]. Other work in the journal applies federated learning to 6G slice resource allocation [14] and multi-agent reinforcement learning to device-to-device relay selection [32]. None of these evaluates against a trivial-policy check or replicates its headline result on an independent seed sample. AXIOM instead shares one graph encoder across agents and trains every pending request as its own temporal-difference sample, for the reason Section 4.1 gives.

Several recent works come close on a single axis of this problem. Shao et al. [31] motivate our GAT encoder choice by showing that graph attention strengthens inter-agent cooperation over plain DQN and Advantage Actor-Critic (A2C) baselines. Ahmadi et al. [2] study topology generalisation through graph attention, which complements but does not answer the fixed-topology evaluation-integrity question we ask in Section 5.2. Sulaiman et al. [33] are the closest prior art to our joint slicing-and-admission-control problem. None of these three checks whether its reward-optimal policy could be mistaken for a degenerate one, and none replicates a headline result on an independent seed sample.

Robustness studies share our concern but stop short of it. Tashman and Cherkaoui [35] and Fatehi et al. [9] study robustness to a disruptive condition, as our disruption campaign in Section 5.6 does, yet neither checks whether the reported

robustness metric is trustworthy and neither replicates a finding on an independent sample. That is the gap our independent-seed replication closes, and the need is concrete, since replication overturned an early churn-immunity reading in our earlier simulator. Zhang et al. [42] are the closest periodic-synchronisation precedent, and Yasin et al. [39] are the closest precedent for federated learning with differential privacy. Their threat model differs from ours, since they secure the controller rather than characterise a control policy's own privacy-utility trade-off, which our σ sweep in Section 5.5 reports, and neither adds a correctness-aware check on whether federating or privatising changed decision quality. A further, closely related set [6, 10, 23, 28] combines two of the three pillars, yet like every work above, each evaluates on its own reported outcome metric, computed once, on one seed or run.

Rather than adding a new point in the GAT, MARL, and FL design space, this paper fills that gap with a demonstration on AXIOM, extended to federation, privacy, and disruption. It shows that two checks are both necessary, namely testing a metric's own trustworthiness and replicating a finding on an independent sample. Skipping either changes what a reader may believe the resulting numbers mean.

3 System Model and Problem Formulation

Fig. 2 provides an overview of the system. The AXIOM controller is evaluated offline in two shared-budget simulation environments, Env A and Env B. Evaluation safeguards and a pre-registered analysis then check every reported result. The deployment pipeline is then checked on a live single-gNB testbed, which Supplementary Section S9 describes.

We consider a cluster of N=3 gNBs, each serving the same three slices, URLLC, eMBB, and mMTC, under a shared per-gNB Physical Resource Block (PRB) budget B. An admission decision is a PRB-ceiling adjustment through the `slicing_control_m` minimum and maximum ratio, rather than a literal per-flow accept or reject, since

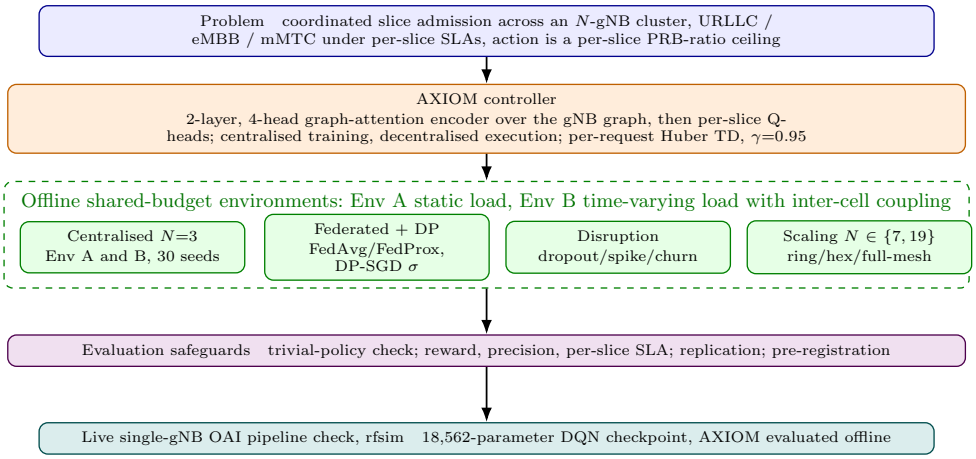


Fig. 2 Overview of the research. An AXIOM controller is evaluated across four offline campaigns in two shared-budget environments, every result checked by evaluation safeguards, with the deployment pipeline checked on a live single-gNB testbed.

the real OAI xApp control interface exposes only ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, as a modelling choice rather than a measured fact. At each control step, every gNB holds zero or more pending admission requests, each tagged with a slice identity, and an agent decides whether to accept or reject for each request independently, conditioned on the current cluster state. The discrete action space matches our single-gNB work exactly at `action_dim`=2, so any coordination benefit found here is due to the architecture and not a richer action space.

The step reward follows our prior single-gNB formulation. It has a per-slice service term net of SLA-violation penalties, a congestion charge on admitted volume, and a load-imbalance penalty across gNBs. The step reward r_t sums over slices indexed by k. It uses priority weight ω_k, per-accept reward R_k, admitted count n_k, violation penalty λ_k, and the indicator $\mathbf{1}[\cdot]$ equal to one exactly when slice k is in SLA violation that step,

$$r_t = \sum_k \big(\omega_k R_k\, n_k - \lambda_k\, \mathbf{1}[v_k]\big) \;-\; \kappa\, C_t \|a_t\|_1 \;-\; \beta\,(1 - F_t), \tag{1}$$

where $\|a_t\|_1$ is the total admitted this step, C_t the cluster-mean congestion level, κ the congestion coefficient, $F_t \in [0, 1]$ the min-over-max PRB-utilisation fairness ratio, and β the load-balance weight. The violation flag v_k follows from a continuous SLA margin m_k. The correctness-aware metrics of Section 4.4 also build on this margin. We take it as the worse of the queue and loss channels,

$$m_k = \min\Big(1 - \tilde{q}_k,\; 1 - \tfrac{\ell_k}{\ell_k^{\max}}\Big), \qquad v_k \Leftrightarrow m_k \le 0, \tag{2}$$

with $\tilde{q}_k$ the slice backlog relative to its budget, and ℓ_k and $\ell_k^{\max}$ its loss proxy and loss budget. The weights are set so that, under congestion, the intended policy is differentiated shedding, which keeps accepting URLLC and eMBB and sheds mMTC.

Every result uses an offline, closed-loop simulation of the cluster, except the bounded live anchor of Section 5.9. Each gNB has a budget of B=100 PRBs shared by its three slices. Each slice's mean offered demand is 1.05 times its nominal PRB share of 30, 60, and 10 PRBs for URLLC, eMBB, and mMTC. Aggregate demand therefore exceeds the budget by about five per cent, while URLLC and eMBB demand stays below their own caps of 33 and 66 PRBs. Whenever the slices' admitted demand exceeds B, they share B in proportion to that demand, so every admitted mMTC request takes resources from URLLC and eMBB. Unserved demand carries forward as backlog within an episode, which drives the SLA margin of Eq. (2). Every episode starts with empty queues, and per-gNB load multipliers drawn from $[0.9, 1.0]$ keep the cluster heterogeneous. We also use a second, harder environment, Env B, which keeps Env A and adds two mechanisms. First, each gNB's mean demand follows $0.95\,(1 + 0.3\sin(2\pi(s + \phi_i)/60))$ times its nominal shares, where s is the step within the episode and the phase ϕ_i is redrawn for every gNB at every episode, so demand swings between about 0.67 and 1.24 times the budget. Second, a gNB's usable budget shrinks with its neighbours' load on the previous step, $B_i(t) = B\,(1 - 0.15\,\bar{u}_{\mathcal{N}(i)}(t-1))$,

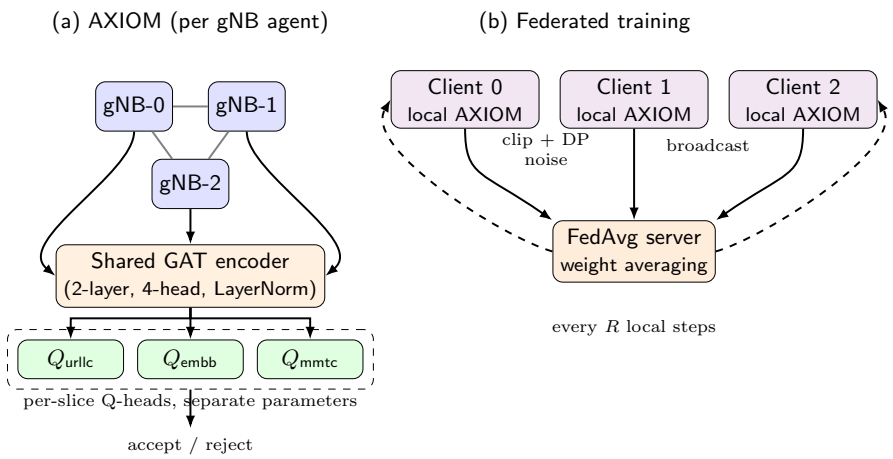


Fig. 3 AXIOM architecture. a, A shared graph-attention encoder over the fully-connected 3-gNB topology produces one embedding per node, consumed at execution time by per-slice Q-heads using only that node's row. b, The federated variant, each gNB a client synchronised by periodic FedAvg with an optional DP-SGD clip-and-noise step.

where $\bar{u}_{\mathcal{N}(i)}$ is the mean PRB utilisation of i's neighbours in the same adjacency graph the encoder reads. This is a first-order model of inter-cell interference, which a gNB observes only through its own queues. The reward weights are unchanged from Env A. In Env B shedding every mMTC request is no longer optimal, since it leaves URLLC compliance at 62.5% at three gNBs, whereas a rule that also sheds eMBB when the URLLC queue builds keeps it at 98.9%. Both constants were calibrated on pilot seeds with rules only, before any reported seed was run. Per-slice compliance counts a slice as compliant only when every gNB meets it, so in Env B it saturates at nineteen gNBs. Env B is therefore run at three and seven gNBs, and the nineteen-gNB evidence comes from Env A.

4 The AXIOM Architecture

AXIOM differs from its two DQN baselines in exactly what each policy sees and shares. AXIOM runs one graph-attention encoder over the whole gNB cluster, trains that encoder jointly from every agent's experience, and reads out one dedicated Q-head per slice, so every gNB decides from a shared view of the cluster that the adjacency graph shapes. The single-agent DQN keeps one controller over a flattened joint state, with no per-gNB structure and no graph. The independent DQN gives each gNB its own learner that sees only local features and never shares parameters or reads the cluster topology. The two baselines are therefore ablations of AXIOM that remove the shared encoder and the graph input together, so any gap between them and AXIOM measures what those components add jointly. A third variant that keeps everything except attention isolates the graph's own contribution in Section 5.3.

Algorithm 1 AXIOM: centralised training, decentralised execution

Require: encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}$, targets $f_{\theta^-}, \{Q_{\phi_k^-}\}$, buffer $\mathcal{D}$, adjacency A, discount γ, sync period C

for each step t, observing joint node features X_t **do**
  $E_t \leftarrow f_\theta(X_t, A)$ {centralised, one row per gNB}
  for each pending request at gNB i, slice k, pick $a \leftarrow \arg\max_a Q_{\phi_k}([E_t[i], X_t[i]])[a]$, ϵ-greedy {decentralised, own row only}
  step env, store $(X_t, X_{t+1}, \{(i,k,a)\}, r_t, \text{done})$ in $\mathcal{D}$
  if $|\mathcal{D}| \geq$ warmup **then**
    sample minibatch B; $E_B \leftarrow f_\theta(X_B, A)$, $E'_B \leftarrow f_{\theta^-}(X'_B, A)$
    per request, $y \leftarrow r + \gamma(1-\text{done}) \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
    $\mathcal{L} \leftarrow$ mean per-request Huber loss, never summed per step; update $\theta, \{\phi_k\}$, clip grad-norm 1.0
  end if
  every C episodes, $\theta^- \leftarrow \theta$, $\phi_k^- \leftarrow \phi_k$
end for

4.1 Shared Encoder, Decentralised Q-Heads

Fig. 3a shows the architecture. At every control step, each gNB forms its own per-slice feature row x_i. The three features are PRB-used-ratio, congestion-level, and queue-length-norm. These rows stack into a joint node-feature matrix $X \in \mathbb{R}^{N \times d}$ over the N gNB nodes, with d features each. A two-layer, H=4-head GAT [38] produces one embedding per gNB. Each head h decomposes additive attention rather than scoring a concatenation. It uses a learned source vector a_s^h and destination vector a_d^h, applied to the shared linear projection W^h. Here ϕ is the leaky rectifier of slope 0.2, and $\mathcal{N}_i$ is the set of neighbours of node i under the adjacency mask,

$$\varepsilon_{ij}^h = \phi\big(a_s^{h\top} W^h x_i + a_d^{h\top} W^h x_j\big), \qquad \alpha_{ij}^h = \frac{\exp \varepsilon_{ij}^h}{\sum_{l \in \mathcal{N}_i} \exp \varepsilon_{il}^h}. \tag{3}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h is its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an ELU nonlinearity, while the final layer averages them. A per-layer normalisation operator $\mathrm{LN}(\cdot)$ then rescales each node's embedding to zero mean and unit variance, and produces the final per-gNB embedding z_i,

$$z_i = \mathrm{LN}\Big(\tfrac{1}{H} \textstyle\sum_{h=1}^{H} \sum_{j \in \mathcal{N}_i} \alpha_{ij}^h W^h x_j\Big). \tag{4}$$

Centralisation comes from this encoder, whose parameters every agent's loss updates jointly, so it sees the full cluster state during training. Decentralised execution comes next, since at action-selection time each agent consumes only its own embedding row z_i to decide its own pending requests.

The `LayerNorm` and the per-slice heads are deliberate design choices. With a single shared Q-head, all three slices update the same parameters, and URLLC's large

priority weight produces gradients that swamp mMTC's much smaller signal, a standard shared-parameter class imbalance. One small multi-layer perceptron (MLP) per slice, selected by the context one-hot, gives each slice its own parameters, and the `LayerNorm` stops an embedding from encoding congestion through its magnitude alone. That normalisation also hides the absolute queue and congestion levels on which a state-dependent admission rule depends, so each node's raw features are concatenated onto its normalised embedding before the Q-heads, $Q_{\phi_k}([z_i, x_i])$, and below we write z_i for that concatenation. This skip connection, together with a learning rate of 10^{-3} in place of the earlier 10^{-4}, is the only change from the original AXIOM design. Both were chosen on pilot seeds of the time-varying environment of Section 3, where the earlier AXIOM learned only the static shedding rule, and both were fixed before any reported seed was run. A seed here is one independent random initialisation of network weights and environment stream, and we report every seed rather than a single best run.

Algorithm 1 states the training loop, where the encoder updates from every request in a sampled minibatch regardless of gNB or slice. This is centralised training, while action selection at rollout time consults only the acting agent's own embedding row, which is decentralised execution. Every pending request at gNB i, slice k is its own temporal-difference sample, and we never sum requests across a step before the loss. For a request with immediate reward r, next-state embedding z'_i, and candidate next action a', the bootstrapped target y at discount γ and the minibatch loss are

$$y = r + \gamma\,(1-d)\max_{a'} Q_{\phi_k^-}(z'_i)[a'], \qquad \mathcal{L} = \frac{1}{|B|}\sum_{(i,k,a)\in B} \ell_\delta\Big(Q_{\phi_k}(z_i)[a] - y\Big), \tag{5}$$

where Q_{ϕ_k} is slice k's online Q-head, $Q_{\phi_k^-}$ its target copy, d the episode-done flag, and B the sampled minibatch of requests. We use the Huber loss ℓ_δ in place of squared error, which keeps the deeper encoder stable under long bootstraps. An earlier variant passed each gNB's summed per-request values through a QMIX-style monotonic mixer [29]. Because the number of pending requests changes from step to step, that mixed target changed scale with the request count, and the mean training loss diverged, rising from about 76 over the first fifth of training steps to about 409 over the last fifth. Treating each request as its own sample removes that dependence.

4.2 Federated Training with Differential Privacy

Fig. 3b shows the federated variant. In this variant, each gNB becomes a federated-learning client. It holds a private copy of the same network and trains only on its own requests. Transitions are partitioned by owning agent, so no client sees another's raw admission data. Each client's GAT encoder still consumes the full joint node-feature snapshot. Every R local steps, the server averages all clients' online weights by FedAvg [21] and broadcasts the result back. We also implement FedProx [19], which adds a proximal term that pulls local weights toward the last broadcast global model. We sweep it at $\mu \in \{0.01, 0.1, 1.0\}$ on three seeds. FedProx gives no consistent improvement over FedAvg, as Section 5.5 reports, so we keep FedAvg.

Differential privacy (DP) bounds how much any one client's training data can change what the shared model reveals. We apply it as a differentially private stochastic gradient descent (DP-SGD) step [1]. Each client clips its minibatch gradient to a fixed norm C_{dp} and, when privacy is enabled, adds Gaussian noise of standard deviation σC_{dp} before its optimiser step. The clip applies to the whole minibatch gradient rather than to each example, so changing any of the client's data in one step moves the clipped gradient by at most $2C_{\mathrm{dp}}$, and the guarantee covers the client's whole step batch. We account for it with zero-concentrated differential privacy (zCDP) [7], where ε and δ are the standard privacy-loss and failure-probability parameters. Composing the Gaussian mechanism over the m noisy steps a client takes gives

$$\rho = \frac{2m}{\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{6}$$

We credit no amplification by subsampling, because minibatches come from a growing replay buffer that reuses records, so Eq. (6) is an upper bound. The clip is applied whether or not noise is added, so a $\sigma{=}0$ run and a $\sigma{>}0$ run differ only in the injected noise. The noise protects each client's own admission decisions and rewards, but not its congestion features, which every client's encoder reads from the shared node-feature snapshot by design. Each federated round therefore runs R local Huber-TD steps per client with an optional FedProx proximal term $\frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$, clips and optionally perturbs the gradient, and closes with a FedAvg average that is broadcast back.

4.3 Experimental Setup

Every campaign uses 300 training and 50 evaluation episodes per seed, and each seed is evaluated on a held-out environment stream. The three-gNB campaigns use 30 seeds, 900 to 929, in each environment, with a disjoint replication sample of seeds 1000 to 1009. The scaling campaigns use 15 seeds per cell at seven gNBs and 30 per cell at nineteen. We compare AXIOM with three learned baselines and several rules. The single-agent DQN is one controller over the flattened joint state. The independent DQN gives each gNB its own DQN with no parameter sharing and no topology information. Both DQN policies are ablations, deliberately reduced variants with a component removed so that its effect can be isolated. MAPPO [40] is a published multi-agent baseline with a shared actor over each gNB's own features and a centralised critic over the joint state. It trains on-policy with proximal policy optimisation, for the same 300 episodes and on the same global reward. Its learning rate, epochs, and minibatch size were tuned on pilot seeds only, because untuned it collapsed into accepting every request in four of five pilot seeds.

Rules need no learning, and accept-all simply admits every request. The oracle rule of Env A sheds mMTC whenever its gNB's congestion level reaches a threshold τ, chosen on training streams only, and the selected value is $\tau{=}0$, so the rule sheds every mMTC request. In Env B we add two tuned rule families, each chosen on training streams over a grid of thresholds and evaluated on the held-out streams. The local family sheds mMTC on the gNB's own congestion or priority-slice queue and sheds eMBB on its URLLC queue, while the neighbour-aware family also reads the mean

congestion of the gNB's neighbours. Two further arms isolate AXIOM's components, namely the DQN baselines at the lower learning rate of 10^{-4}, and an AXIOM variant whose graph encoder is replaced by a per-node MLP of the same width that ignores the adjacency.

All learned arms share the discount factor $\gamma = 0.95$, the training budget, and the exploration schedule, and AXIOM and the DQN baselines train at learning rate 10^{-3}. Supplementary Section S10 lists every hyperparameter and the design of the two sensitivity studies, which change the load and the slice weights in Env A and the learning rate and encoder width in Env B.

All confidence intervals (CIs) are 95% bootstrap percentile intervals over 10,000 resamples. Paired comparisons use the Wilcoxon signed-rank test, and each family of comparisons against AXIOM is Holm-corrected for multiple testing.

The settings, seeds, and hypotheses of the final campaign were fixed in a dated pre-registration, committed before any reported seed was run. It includes the change to AXIOM's learning rate and skip connection and the constants of Env B, both chosen on pilot seeds 2000 to 2009. It states four hypotheses, all fixed before any result. H1 is that AXIOM beats both trivial rules in Env B. H2 covers how AXIOM compares with every learned baseline and tuned rule in Env B. H3 asks whether graph attention beats the per-node encoder in Env B at seven gNBs. H4 covers how AXIOM compares with every baseline at nineteen gNBs in Env A, over 30 seeds per layout.

A pre-registered validation screen checks every trained checkpoint of every learned arm on its own training stream before evaluation, and every seed is reported whether or not it passes, with the screen's criteria in Supplementary Section S10.

4.4 Evaluation Safeguards

Three safeguards make the results below trustworthy, and Table 1 lists the failure each one caught during this work. A standard pipeline would have reported every one of them as a clean result.

The first is a trivial-policy check, in which we evaluate the simplest rules the environment admits, accepting every request and shedding one slice outright, before trusting any learned policy. It exposed an earlier simulator in which accepting every request was optimal, and the shared-budget simulator of Section 3 removes that loophole.

The second is a correctness-aware set of metrics, computed from data every run already logs. Mean reward per step is the training objective. Block precision is the fraction of blocked requests that correctly target mMTC, the slice the reward calibration makes reject-optimal. Writing b_{mMTC} and b for the mMTC-targeted and total blocks over an evaluation, it is $\pi = b_{\mathrm{mMTC}}/b$ for $b > 0$ and undefined for $b = 0$, so a seed that never blocks is counted as a collapse rather than scored as a success. Per-slice SLA compliance then shows whether the shedding actually protects URLLC and eMBB. We do not use the common all-slice compliance metric, because correct shedding gives up mMTC's own SLA by design, so that metric scores a correct policy and a failed one alike.

The third is a reproduction-versus-replication discipline, in which we recompute every headline comparison on an independent, disjoint seed sample and report both.

Table 1 Evaluation failures this paper's safeguards caught during the work, none of which a standard pipeline would have surfaced.

Failure	Missed by	Caught by	Outcome
Inert simulator (§3)	Learned-policy metrics	Accept-all baseline	Shared-budget rerun
No-op scored as fine	All-slice compliance	Precision, per-slice SLA	Collapse made visible
Repro. gap (Suppl. S12)	Single-sample claim	Repro-vs-replication	Churn claim retracted
Backlog carry-over (§4.4)	Per-seed means	Replication, step traces	Episode reset, rerun
Static AXIOM in Env B (§4.1)	Env A results	Env B pilot seeds	Faster learning rate, skip
Screen miscalibrated (§4.3)	Env A screen design	Pilot screen check	Pre-registered amendment
Scheduler floor (§5.9)	Cross-layer floor	Direct instrumentation	100k+ readings clear

It caught queue backlog that carried over between episodes, and every result in this paper comes from the corrected simulator, which empties all queues at the start of each episode.

5 Results

We compare AXIOM with three learned baselines and a family of rules in two environments. AXIOM is the architecture of this paper, a shared encoder that reads the whole base-station cluster and feeds one small decision head per slice. The single-agent DQN flattens the cluster into one controller that sees no per-gNB structure. The independent DQN gives each gNB its own learner that never looks at its neighbours. MAPPO shares one actor across gNBs and trains it with a centralised critic. The rules range from accept-all and static mMTC shedding, the optimum of Env A, to threshold rules tuned on training streams.

The comparison rests on four quantities, the first being mean reward per step, the objective the policies train on, so higher is better. Block precision is the share of a policy's rejections that land on mMTC. Per-slice SLA compliance shows whether a policy's shedding actually protects URLLC and eMBB, and we summarise it as priority-slice compliance, the mean of the two. Collapse names a failure mode in which a trained policy stops discriminating and simply accepts every request. Table 2 sets out how the four learned policies stand, and the subsections that follow give the evidence.

AXIOM's clearest advantage appears as the cluster grows. In Env A it keeps priority-slice compliance near 97% at seven and nineteen gNBs while every baseline falls to between 56% and 76%, with a parameter count that does not grow with the cluster. It also leads every learned baseline in Env B at three gNBs. Its advantage has limits, since a tuned threshold rule beats every learned policy in Env B and the single-agent DQN keeps pace with AXIOM in Env A at three gNBs and in Env B at seven.

5.1 Static Load, Three gNBs

Table 3 and Fig. 4 report the thirty-seed campaign in Env A. AXIOM never collapses, 99.2% of its blocks land on mMTC, and it keeps URLLC and eMBB at 99.8% and 97.9% SLA compliance. Its mean reward of 12.698 per step is 98.9% of the oracle rule's

12.835, and the two tie exactly in 22 of the 30 seeds. AXIOM leads the independent DQN by +1.110 per step, with 95% CI [0.60, 1.65], and MAPPO by +2.685, with 95% CI [2.01, 3.35], both at Holm-adjusted $p \leq 0.0002$. Its lead over the single-agent DQN, +0.311 with 95% CI [−0.01, 0.64], does not survive correction on thirty seeds, at $p = 0.066$. MAPPO trails both DQN baselines here, because it sheds eMBB as well as mMTC, which lets eMBB compliance fall to 57.6%.

A disjoint replication on seeds 1000 to 1009 is stronger for AXIOM. There it leads the single-agent DQN, the independent DQN, and MAPPO by +1.083, +0.799, and +3.806 per step at Holm-adjusted $p \leq 0.023$, while sitting only 0.018 below the oracle. Pooled over all 40 seeds, AXIOM leads the three baselines by +0.504, +1.032, and

Table 2 How the four learned policies compare. Env A has static load, Env B time-varying load with inter-cell coupling. Priority SLA is the mean of URLLC and eMBB compliance.

Property	AXIOM	Single-agent DQN	Independent DQN	MAPPO
Design	Shared encoder, per-slice heads	One flattened controller	One learner per gNB	Shared actor, central critic
Parameters at execution	7,782 at any size	21,250 to 41,730	55,302 to 350,246	5,122
Env A reward, 3 gNBs	12.70, 98.9% of oracle	12.39	11.59	10.01
Env A priority SLA, 7 and 19 gNBs	97.1% and 97.3%	76.2% and 65.3%	74.6% and 56.4%	67.7% and 60.6%
Collapse, Env A, 19 gNBs	1/90	84/90	3/90	0/90
Env B reward, 3 gNBs	8.83	7.49	7.79	6.97
Env B priority SLA, 7 gNBs	35.1%	44.3%	39.9%	35.9%
Federated, noise-free	No significant cost	Not applicable	Not applicable	Not applicable

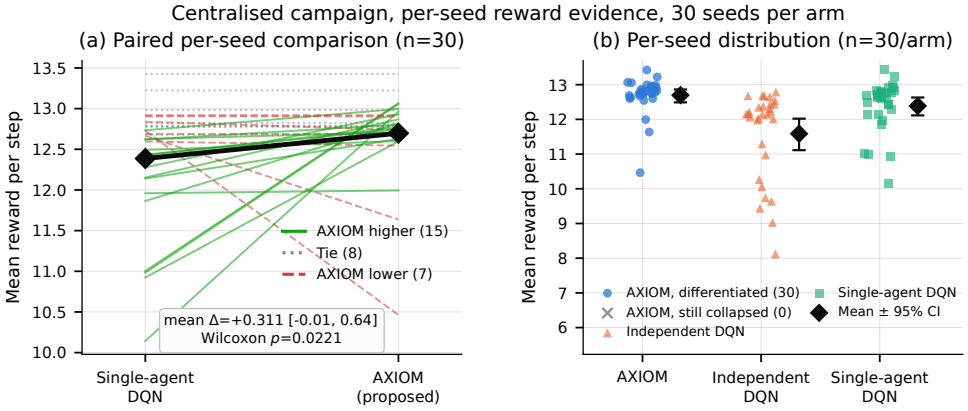


Fig. 4 Env A, three gNBs, per-seed evidence. a, Paired per-seed mean reward per step of AXIOM versus single-agent DQN, coloured by win, tie, or loss. b, Per-seed reward distribution for AXIOM and the two DQN baselines.

Table 3 Env A, three gNBs, 30 seeds per arm, 95% bootstrap CIs. Precision averages the seeds that block at all. SLA columns give per-slice compliance.

Arm	Reward/step	Precision	Collapsed	URLLC	eMBB
Oracle rule	12.835 [12.77, 12.91]	1.000	0/30	1.000	1.000
AXIOM	12.698 [12.49, 12.86]	0.992	0/30	0.998	0.979
AXIOM, MLP encoder	12.640 [12.46, 12.79]	0.986	0/30	1.000	0.976
Single-agent DQN	12.387 [12.11, 12.63]	0.990	0/30	0.994	0.941
at LR 10^{-4}	11.477 [10.61, 12.09]	0.944	0/30	0.979	0.866
Independent DQN	11.587 [11.11, 12.02]	0.962	0/30	0.993	0.865
at LR 10^{-4}	11.119 [10.73, 11.50]	0.878	0/30	0.985	0.794
MAPPO	10.013 [9.30, 10.73]	0.707	2/30	0.978	0.576

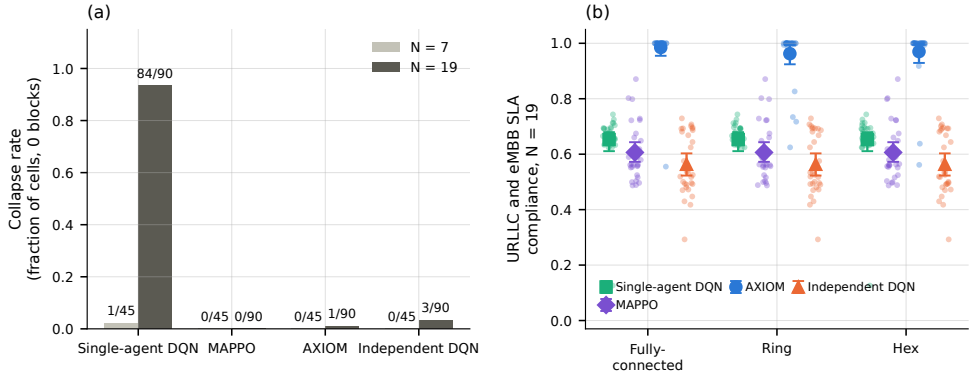


Fig. 5 Env A scaling. a, Collapse rate pooled over layouts, with collapsed over total cells. b, Mean URLLC and eMBB compliance at 19 gNBs by layout, 30 seeds per cell, per-seed points behind the mean and 95% CI.

+2.965 at Holm-adjusted $p \leq 0.002$, and falls 0.108 short of the oracle, a small gap that is significant at $p = 0.034$. Fresh seeds in Section 5.8 do not reproduce the lead over the single-agent DQN, however.

5.2 Cluster-Size Scaling

We grow the cluster in Env A from three gNBs to seven and nineteen under three layouts, fully-connected, ring, and hex-grid. Every gNB keeps its own shared 100-PRB budget and its own demand, so per-gNB load stays the same as the cluster grows. Each cell runs 15 seeds at seven gNBs and 30 at nineteen, where the earlier comparison had been underpowered. The independent DQN ignores the layout, so it gives identical results under all three layouts, which checks that the layout changes nothing else. The single-agent DQN also ignores the layout, yet its results vary between layouts in some seeds, because its training is not bit-reproducible, for the reason Section 5.8 gives. We test the per-layout gaps on reward and on priority-slice compliance, Holm-correcting the 18 comparisons at each size.

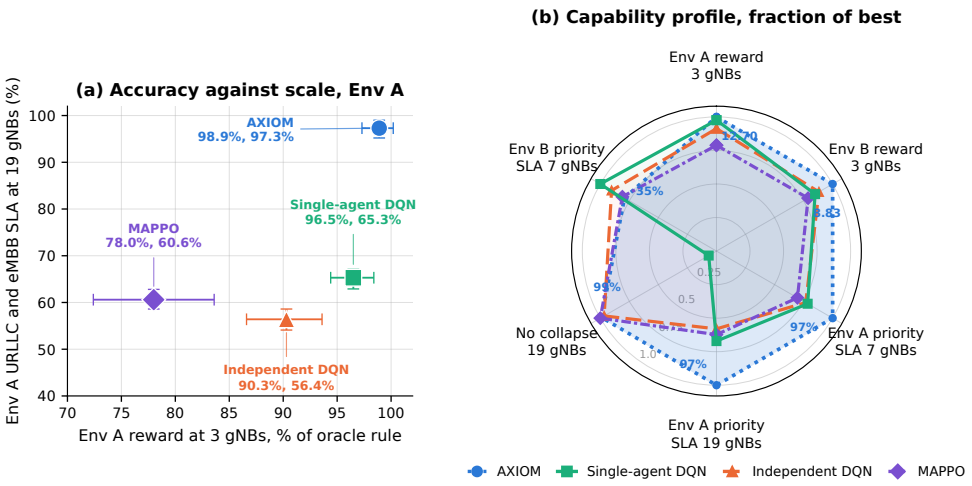


Fig. 6 AXIOM against the three learned baselines. a, Env A reward at three gNBs as a share of the oracle rule against compliance at 19 gNBs, with 95% CIs. b, Six measured axes, each as a fraction of the best learned policy on it.

At seven gNBs AXIOM keeps URLLC and eMBB at 99.8% and 94.4%, against 91.4% and 61.0% for the single-agent DQN, 94.2% and 55.0% for the independent DQN, and 87.0% and 48.4% for MAPPO. On fresh seeds the two DQN baselines reach up to 83.3%, as Section 5.8 reports. Under every layout it leads all three baselines on priority-slice compliance by 17.9 to 29.9 percentage points and on cluster reward by 3.40 to 7.07 per step. All 18 comparisons hold after Holm correction, at adjusted $p \leq 0.011$.

At nineteen gNBs the gaps widen, and they answer hypothesis H4. AXIOM keeps URLLC and eMBB at 99.4% and 95.1%, a pooled priority-slice compliance of 97.3% with 95% CI [95.2%, 99.0%]. The single-agent DQN, the independent DQN, and MAPPO keep 65.3%, 56.4%, and 60.6%. Under every layout AXIOM leads every baseline by 31.0 to 42.1 percentage points of compliance and by 10.7 to 18.6 reward per step, winning 28 to 30 of 30 seeds. All 18 comparisons hold after Holm correction, at adjusted $p \leq 0.0001$. Fig. 5a shows the single-agent DQN collapsing into accepting every request in 84 of its 90 cells, against 3 for the independent DQN, 1 for AXIOM, and none for MAPPO. The original AXIOM kept priority-slice compliance at only 79.1% at this size, so the faster learning rate matters most at scale.

Fig. 6a places the four learned policies on one map of accuracy at three gNBs against priority-slice protection at nineteen, in Env A. AXIOM sits highest on both axes, at 98.9% of the oracle reward and 97.3% compliance. The single-agent DQN reaches 96.5% and 65.3%, the independent DQN 90.3% and 56.4%, and MAPPO 78.0% and 60.6%. The radar in Fig. 6b adds Env B reward at three gNBs, collapse resistance, and Env B compliance at seven gNBs. AXIOM leads on five of the six axes shown. The exception is Env B compliance at seven gNBs, where the single-agent DQN leads.

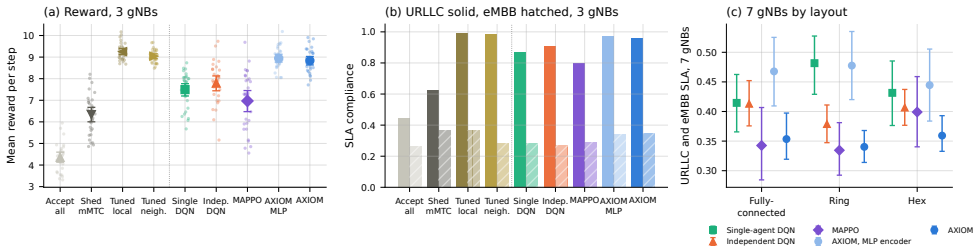


Fig. 7 Env B. a, Reward at three gNBs, 30 seeds, per-seed points behind the mean and 95% CI. b, URLLC and eMBB compliance at three gNBs. c, Priority-slice compliance of the learned arms at seven gNBs by layout, 15 seeds per cell.

5.3 Ablations and Sensitivity

Three checks test where AXIOM's behaviour comes from. First, the learning rate matters for AXIOM and its baselines alike. In Env B the original AXIOM, at 10^{-4}, earns 2.37 less per step than the current design in all ten seeds, while a faster rate or a wider encoder changes nothing significant, and Supplementary Section S12 gives the full study.

Second, graph attention adds nothing measurable anywhere we tested it. In Env A the per-node MLP variant ties AXIOM at three gNBs, with a difference of +0.058 and 95% CI [−0.17, 0.26], and at nineteen gNBs on the ring layout, +0.31 per step with 7 of 12 seeds tied. Env B's inter-cell coupling gives neighbour information real value, yet the pre-registered test H3 at seven gNBs even leans towards the MLP. The MLP keeps 1.12 and 0.95 more reward per step on the ring and hex layouts, winning 10 of 15 seeds in each, although neither difference survives Holm correction, at $p = 0.061$ and $p = 0.095$. AXIOM's gains therefore come from its shared encoder, per-slice heads, and joint training rather than from attention. One explanation is that a gNB's own queues already reflect most of what its neighbours do to it, so attending to them adds little. Following the pre-registration, the title of this paper therefore no longer names graph attention.

Third, AXIOM's behaviour holds in Env A when the environment changes. Under three changed conditions, namely load at 1.02 and 1.07 times the nominal shares and the changed slice weights, AXIOM stays within 0.13 of the oracle rule's reward with no significant difference. On fresh seeds the gap at 1.02 times the load was larger in a ten-seed sample but not in a twenty-seed one, as Section 5.8 reports. On these seeds it keeps a positive lead over both DQN baselines under every condition, but at ten seeds only two of those six leads survive correction. They are its lead over the independent DQN at 1.07 times the load, +1.073 at $p = 0.006$, and over the single-agent DQN under the changed weights, +1.264 at $p = 0.041$.

5.4 Time-Varying Load with Inter-Cell Coupling

Env B tests the case a static benchmark cannot, where no fixed rule is optimal. Table 4 and Fig. 7 report the thirty-seed campaign at three gNBs and fifteen seeds per layout

at seven. First, adapting to the load pays off clearly. Shedding every mMTC request, the optimum of Env A, now leaves URLLC compliance at 62.5%. AXIOM beats that rule by +2.495 reward per step and accept-all by +4.476, winning all 30 seeds against each at Holm-adjusted $p < 0.0001$, as hypothesis H1 predicted. AXIOM keeps URLLC at 95.8% by shedding eMBB as well as mMTC when URLLC queues build.

Second, among learned policies AXIOM leads at three gNBs. It beats the independent DQN, the single-agent DQN, and MAPPO by +1.038, +1.337, and +1.863 reward per step and also on priority-slice compliance, winning 24 to 29 of 30 seeds, all at Holm-adjusted $p < 0.0001$. The disjoint ten-seed replication points the same way, with AXIOM winning 7 or 8 of 10 seeds against each, but no lead survives correction there.

Third, a tuned rule beats every learned policy. The best local rule sheds every mMTC request and sheds eMBB whenever the URLLC queue passes 0.2, and it earns 9.252 per step, 0.420 more than AXIOM, winning 22 of 30 seeds at $p < 0.0001$. The neighbour-aware rule earns slightly less, at 9.041, so neighbour information did not help the rules either. The rule encodes domain knowledge of which slice to shed under which pressure, which AXIOM must learn from the reward alone in 300 episodes.

At seven gNBs every learned policy falls further behind. The tuned rules earn 26.5 to 27.3 per step against 20.3 to 24.6 for the learned policies, and they beat AXIOM in all 15 seeds under every layout. AXIOM itself earns 23.4 to 23.6, no more than the static rule's 23.6, and keeps priority-slice compliance at 34% to 36%, close to the static rule. On the ring layout that is below the single-agent DQN, at Holm-adjusted $p = 0.011$. AXIOM's only significant leads over a learned baseline at this size are over MAPPO's reward, under the fully-connected and ring layouts. In Env B, learned multi-agent admission control therefore beats a static rule only at three gNBs, and never beats a rule tuned with domain knowledge.

5.5 Federated Training and DP

With federation switched on and no noise, mean reward per step on the ten shared seeds is 12.356 for centralised AXIOM and 11.929 for the federated variant. That is a difference of +0.427 with 95% CI [−0.54, 1.51] and $p = 0.65$, so federation shows no significant cost, although it is more variable, with five seeds losing and four gaining.

Table 4 Env B, three gNBs, 30 seeds per arm, 95% bootstrap CIs. Rules above the line, learned policies below it.

Arm	Reward/step	URLLC	eMBB	mMTC
Accept all	4.356 [4.13, 4.60]	0.446	0.265	0.955
Shed all mMTC	6.337 [6.00, 6.67]	0.625	0.367	0.371
Tuned local rule	9.252 [9.12, 9.40]	0.989	0.367	0.390
Tuned neighbour-aware rule	9.041 [8.93, 9.15]	0.988	0.283	0.901
AXIOM	8.832 [8.64, 9.03]	0.958	0.344	0.406
AXIOM, MLP encoder	8.948 [8.77, 9.13]	0.974	0.337	0.452
Single-agent DQN	7.495 [7.20, 7.77]	0.866	0.284	0.412
Independent DQN	7.794 [7.44, 8.13]	0.910	0.270	0.441
MAPPO	6.968 [6.47, 7.45]	0.797	0.289	0.579

The disjoint replication shows a larger cost of +0.968, with 95% CI [0.18, 1.96], which falls just short of significance at $p = 0.055$.

Adding DP-SGD noise to every step lowers reward by 0.77 to 1.25 per step, and no drop survives Holm correction at ten seeds. The formal guarantee is weak at these settings, with ε of about 2,600 or more, so per-step DP-SGD delivers no meaningful guarantee here, and Supplementary Section S12 reports the privacy analysis in full.

5.6 Disruption

We inject three mid-episode disruptions, gNB dropout, demand spikes, and agent churn, into every policy's frozen Env A checkpoint without retraining. Dropout and churn hurt every policy significantly at every severity, and no gap between AXIOM and any baseline survives Holm correction under any disruption. Demand spikes leave AXIOM's decisions largely intact, with block precision at 0.98, and Supplementary Section S12 gives the protocol and every result.

5.7 Model Complexity

Table 5 sets out the cost. AXIOM's shared encoder keeps its parameter count at 7,782 at every cluster size, whereas the independent DQN grows from 55,302 to 350,246 parameters and the single-agent DQN's input grows with the cluster. AXIOM's greedy decisions take 1.0 to 4.8 ms per control step on one CPU thread, well inside the 5 s control period of the live testbed. MAPPO is fastest at inference, because it scores all of a step's requests in one batch.

5.8 Reproducibility and Fresh-Seed Verification

We checked the results in three ways, with every pass criterion fixed before the run it judged. First, rerunning eleven reported cells on their original seeds in fresh processes reproduced every stored per-seed result exactly, with one exception. The single-agent DQN's training code builds its network before setting the seed, so the first seed of each batch starts from an unseeded initialisation. Rerunning this baseline with its initialisation seeded leaves AXIOM's leads in Env B and at nineteen gNBs unchanged.

Second, four further runs retrained every arm on fresh seeds, 10 to 40 per cell, and 81 of their 90 pre-stated checks held. At seven and nineteen gNBs in Env A AXIOM kept 93.4% to 100% priority-slice compliance and beat every baseline under every layout, at Holm-adjusted $p \leq 0.032$ in all but one ten-seed cell, which a twenty-seed

Table 5 Model complexity. Parameters used at execution, greedy inference time per control step on one CPU thread, and median training time per seed.

Policy	Parameters, 3/7/19 gNBs	ms per step, 3/7/19 gNBs	Training s, 3/19 gNBs
AXIOM	7,782 / 7,782 / 7,782	1.04 / 1.98 / 4.79	624 / 892
AXIOM, MLP encoder	3,190 / 3,190 / 3,190	0.77 / 1.69 / 4.46	503 / –
Single-agent DQN	21,250 / 26,370 / 41,730	0.55 / 1.30 / 3.58	85 / 202
Independent DQN	55,302 / 129,038 / 350,246	0.59 / 1.38 / 3.73	181 / 1054
MAPPO	5,122 / 5,122 / 5,122	0.19 / 0.20 / 0.22	73 / 256

rerun then confirmed at $p \leq 0.007$. In Env B at three gNBs it again beat the static rule and every learned baseline at Holm-adjusted $p \leq 0.0002$, the tuned rules again beat every learned policy, and the disruption, federation, and privacy results held as reported. The Supplementary Material reports every number from these runs.

The nine checks that failed concern secondary margins rather than any headline result. On fresh seeds the DQN baselines reached up to 83.3% compliance at seven gNBs, above the 80% bound we had set. AXIOM's small lead over the single-agent DQN in Env A at three gNBs did not recur, at -0.021 and $+0.159$ per step on 20 and 40 new seeds, and at 1.02 times the load that baseline matched AXIOM. Two weak AXIOM seeds also left one ten-seed load sample 3.2% short of the oracle, a gap that a twenty-seed rerun did not reproduce. Third, a script written independently of the analysis pipeline, with its own log reader and Holm correction, recomputes from the raw logs the numbers in every results table and the key results sentences, and all 295 of its checks pass.

5.9 Live Deployment

We also checked the deployment path that AXIOM would use on a real single-gNB OAI testbed, running an offline-trained single-agent DQN checkpoint through the real E2 control loop with real UEs and per-slice UDP traffic. The pipeline worked end to end, with 0.000% loss at both loads and zero below-floor grants across more than 100,000 readings after a scheduler-configuration fix. AXIOM itself is evaluated offline, since our two-gNB live attempt did not sustain traffic, and Supplementary Section S9 gives the testbed and every live result.

6 Conclusion

This paper presented AXIOM, a multi-agent slice admission controller with a shared cluster encoder and decentralised per-slice action heads, together with an evaluation protocol designed to make its results trustworthy. It targets three gaps from our prior systematic review. It closes the first, trustworthy evaluation, tests the second, privacy-preserving coordination, and finds it still open, and validates only the control path of the third, live deployment.

AXIOM's clearest advantage appears as the cluster grows. In a static-load shared-budget environment it keeps priority-slice compliance near 97% at seven and nineteen gNBs, while single-agent and independent DQNs and MAPPO stay below 84%, with a parameter count fixed at 7,782, and it reaches 98.9% of an oracle rule's reward at three gNBs. In a harder environment with time-varying load and inter-cell coupling, AXIOM beats the static and accept-all rules and leads every learned baseline at three gNBs, and federated training showed no significant cost. These gains have two limits that we state plainly. A three-parameter rule tuned with domain knowledge still beats every learned policy under coupling, and the single-agent DQN keeps pace with AXIOM in Env A at three gNBs and in Env B at seven. Graph attention added nothing over a per-node encoder, so the gains come from the shared encoder, the per-slice heads, and joint training, and per-step DP-SGD gave only a weak guarantee. The main findings

held on fresh seeds, and the live testbed validates the control path rather than AXIOM itself, which is evaluated offline.

The evaluation safeguards changed what this paper finally reports, exposing an inert earlier simulator, queue backlog carried over between episodes, and an AXIOM design that learned only a static rule, and every reported result comes from the corrected versions. The broader lesson is that learned admission control should be judged against trivial and tuned rules, with metrics tied to the objective, on independent seeds, under a pre-registered analysis.

Future work targets the gaps that this paper leaves open. The first is a live multi-gNB deployment of AXIOM itself. The second is closing the gap to tuned rules in coupled environments, for instance through longer training or by giving the learner the queue-threshold structure the rules exploit. The third is a privacy layer that adds noise to the per-round model updates rather than to every step, so that the formal guarantee becomes meaningful. We also plan Multi-Access Edge Computing (MEC) scaling, which adds compute-placement nodes to the cluster graph, cross-operator federation, and direct Quality of Experience (QoE) inference in the reward.

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Declarations

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- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable, since this study involves no human participants, animal subjects, or identifiable patient data. All reported results are either offline simulation campaigns or measurements from a laboratory radio testbed.
- **Consent for publication:** Not applicable, since this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable, since this study involves no physical or biological materials.
- **Code availability:** The framework and experiment scripts used to produce this paper's results, including the reproducibility manifest and analysis scripts used to independently re-derive every table and figure in this paper, are available from the corresponding author on reasonable request.

- **Author contribution:** M. P. Kunasegran: conceptualisation, methodology, software, investigation, writing (original draft). W. L. Pang: supervision, writing (review and editing). S. K. Phang: supervision, writing (review and editing).

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